

Chapter 6 Orthogonality and Least Squares

Section 6.1: Inner Product, Length, and Orthogonality

Section 6.2: Orthogonal Sets

Section 6.3: Orthogonal Projections

Section 6.4: The Gram-Schmidt Process

Section 6.5: Least-Squares Problems

Inner Product

- The number $\mathbf{u}^T \mathbf{v}$ is called the **inner product** of $\mathbf{u}$ and $\mathbf{v}$, and it is written as $\mathbf{u} \cdot \mathbf{v}$.
- The inner product is also referred to as a **dot product**.

Inner Product

• If $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$, then the inner product of $\mathbf{u}$ and $\mathbf{v}$ is

$$[u_1 \quad u_2 \quad \cdots \quad u_n] \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$$

Inner Product

- If $\mathbf{u}$ and $\mathbf{v}$ are vectors in $\mathbb{R}^n$, then we regard $\mathbf{u}$ and $\mathbf{v}$ as $n \times 1$ matrices.
- The transpose $\mathbf{u}^T$ is a $1 \times n$ matrix, and the matrix product $\mathbf{u}^T \mathbf{v}$ is a 1×1 matrix, which we write as a single real number (a scalar) without brackets.

Inner Product

Theorem 1: Let $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ be vectors in $\mathbb{R}^n$, and let c be a scalar. Then

(a) $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$

(b) $(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$

(c) $(c\mathbf{u}) \cdot \mathbf{v} = c(\mathbf{u} \cdot \mathbf{v}) = \mathbf{u} \cdot (c\mathbf{v})$

(b) $\mathbf{u} \cdot \mathbf{u} \geq 0$, and $\mathbf{u} \cdot \mathbf{u} = 0$ if and only if $\mathbf{u} = \mathbf{0}$

- Properties and can be combined several times to produce the following useful rule:

$$(c_1\mathbf{u}_1 + \cdots + c_p\mathbf{u}_p) \cdot \mathbf{w} = c_1(\mathbf{u}_1 \cdot \mathbf{w}) + \cdots + c_p(\mathbf{u}_p \cdot \mathbf{w})$$

The Length of a Vector

- If $\mathbf{v}$ is in $\mathbb{R}^n$, with entries $v_1, \dots, v_n$, then the square root of $\mathbf{v} \cdot \mathbf{v}$ is defined because $\mathbf{v} \cdot \mathbf{v}$ is nonnegative.

- **Definition:** The **length** (or **norm**) of $\mathbf{v}$ is the nonnegative scalar $\|\mathbf{v}\|$ defined by

$$\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}} = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2} \text{ and } \|\mathbf{v}\|^2 = \mathbf{v} \cdot \mathbf{v}$$

- Suppose $\mathbf{v}$ is in $\mathbb{R}^2$, say, $\mathbf{v} = \begin{bmatrix} a \\ b \end{bmatrix}$.

The Length of a Vector

- If we identify $\mathbf{v}$ with a geometric point in the plane, as usual, then $\|\mathbf{v}\|$ coincides with the standard notion of the length of the line segment from the origin to $\mathbf{v}$.
- This follows from the Pythagorean Theorem applied to a triangle such as the one shown in the following figure.

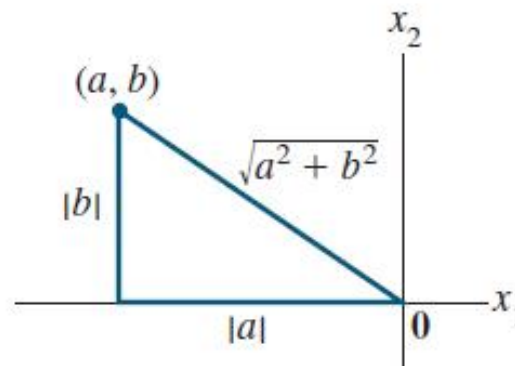


FIGURE 1

Interpretation of $\|\mathbf{v}\|$ as length.

The Length of a Vector

- For any scalar c , the length $c\mathbf{v}$ is $|c|$ times the length of $\mathbf{v}$.
That is, $\|c\mathbf{v}\| = |c|\|\mathbf{v}\|$.
- A vector whose length is 1 is called a **unit vector**.
- If we divide a nonzero vector $\mathbf{v}$ by its length—that is, multiply by $1/\|\mathbf{v}\|$ - we obtain a unit vector $\mathbf{u}$ because the length of $\mathbf{u}$ is $(1/\|\mathbf{v}\|)\|\mathbf{v}\| = 1$.
- The process of creating $\mathbf{u}$ from $\mathbf{v}$ is sometimes called **normalizing $\mathbf{v}$** , and we say that $\mathbf{u}$ is in the same direction as $\mathbf{v}$.

The Length of a Vector

- **Example:** Let $\mathbf{v} = (1, -2, 2, 0)$. Find a unit vector $\mathbf{u}$ in the same direction as $\mathbf{v}$.
- **Solution:** First, compute the length of $\mathbf{v}$:

$$\|\mathbf{v}\|^2 = \mathbf{v} \cdot \mathbf{v} = 9, \|\mathbf{v}\| = 3$$

- Then, multiply $\mathbf{v}$ by $1/\|\mathbf{v}\|$ to obtain

$$\mathbf{u} = \frac{1}{\|\mathbf{v}\|} \mathbf{v} = \frac{1}{3} \mathbf{v} = \frac{1}{3} \begin{bmatrix} 1 \\ -2 \\ 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/3 \\ -2/3 \\ 2/3 \\ 0 \end{bmatrix}$$

Distance in $\mathbb{R}^n$

- To check that $\|\mathbf{u}\| = 1$, it suffices to show that $\|\mathbf{u}\|^2 = 1$.

$$\begin{aligned}\|\mathbf{u}\|^2 &= \mathbf{u} \cdot \mathbf{u} = \left(\frac{1}{3}\right)^2 + \left(-\frac{2}{3}\right)^2 + \left(\frac{2}{3}\right)^2 + (0)^2 \\ &= \frac{1}{9} + \frac{4}{9} + \frac{4}{9} + 0 = 1\end{aligned}$$

- **Definition:** For $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$, the **distance between $\mathbf{u}$ and $\mathbf{v}$** , written as $\text{dist}(\mathbf{u}, \mathbf{v})$, is the length of the vector $\mathbf{u} - \mathbf{v}$. That is,

$$\text{dist}(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$$

Distance in $\mathbb{R}^n$

- **Example:** Compute the distance between the vectors $\mathbf{u} = (7,1)$ and $\mathbf{v} = (3,2)$.
- **Solution:** Calculate

$$\mathbf{u} - \mathbf{v} = \begin{bmatrix} 7 \\ 1 \end{bmatrix} - \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}$$

$$\|\mathbf{u} - \mathbf{v}\| = \sqrt{4^2 + (-1)^2} = \sqrt{17}$$

When the vector $\mathbf{u} - \mathbf{v}$ is added to $\mathbf{v}$, the result is $\mathbf{u}$.

Distance in $\mathbb{R}^n$

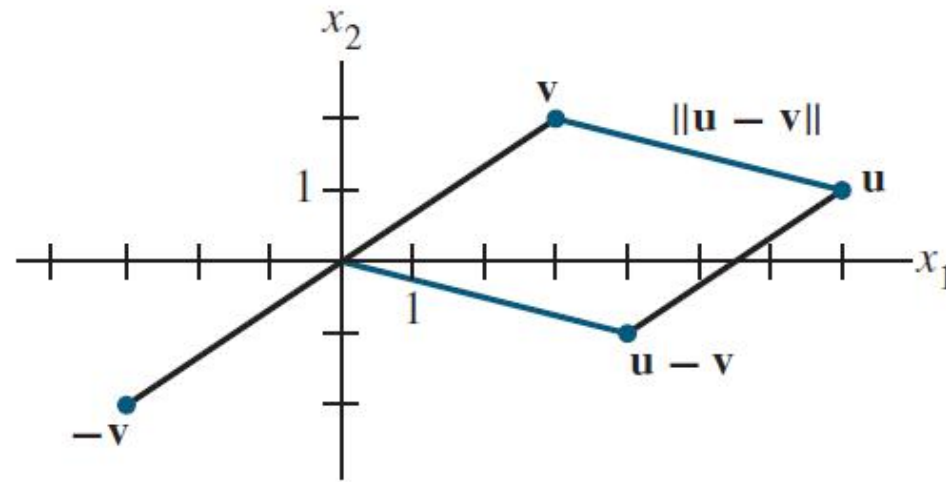
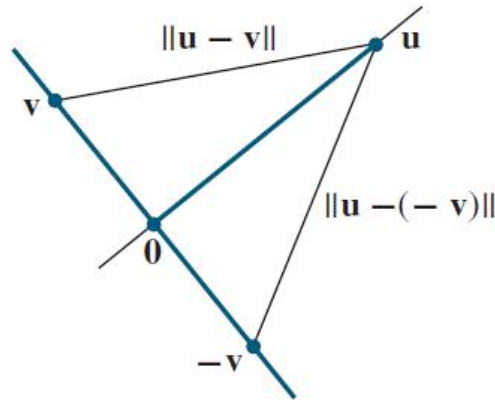


FIGURE 4 The distance between $\mathbf{u}$ and $\mathbf{v}$ is the length of $\mathbf{u} - \mathbf{v}$.

- Notice that the parallelogram in the above figure shows that the distance from $\mathbf{u}$ to $\mathbf{v}$ is the same as the distance from $\mathbf{u} - \mathbf{v}$ to $\mathbf{0}$.

Orthogonal Vectors

- Consider $\mathbb{R}^2$ or $\mathbb{R}^3$ and two lines through the origin determined by vectors $\mathbf{u}$ and $\mathbf{v}$.
- See the figure below. The two lines shown in the figure are geometrically perpendicular if and only if the distance from $\mathbf{u}$ to $\mathbf{v}$ is the same as the distance from $\mathbf{u}$ to $-\mathbf{v}$.



- This is the same as requiring the squares of the distances to be the same.

Orthogonal Vectors

$$\begin{aligned} [\text{dist}(u, -v)]^2 &= \|u - (-v)\|^2 = \|u + v\|^2 & [\text{dist}(u, v)]^2 &= \|u\|^2 + \|-v\|^2 + 2u \cdot (-v) \\ &= (u + v) \cdot (u + v) & &= \|u\|^2 + \|v\|^2 - 2u \cdot v \\ &= u \cdot (u + v) + v \cdot (u + v) \\ &= u \cdot u + u \cdot v + v \cdot u + v \cdot v \\ &= \|u\|^2 + \|v\|^2 + 2u \cdot v \end{aligned}$$

- The two squared distances are equal if and only if $2u \cdot v = -2u \cdot v$, which happens if and only if $u \cdot v = 0$.

Orthogonal Vectors

- **Definition:** Two vectors $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ are **orthogonal** (to each other) if $\mathbf{u} \cdot \mathbf{v} = 0$.
- The zero vector is orthogonal to every vector in $\mathbb{R}^n$ because $\mathbf{0}^T \mathbf{v} = 0$ for all $\mathbf{v}$.

The Pythagorean Theorem

- **Theorem:** Two vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal if and only if

$$\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$$

Orthogonal Complements

- If a vector $\mathbf{z}$ is orthogonal to every vector in a subspace W of $\mathbb{R}^n$, then $\mathbf{z}$ is said to be **orthogonal to W** .
- The set of all vectors $\mathbf{z}$ that are orthogonal to W is called the **orthogonal complement of W** and is denoted by $W^\perp$ (and read as “ W perpendicular” or simply “ W perp”).

Orthogonal Complements

1. A vector $\mathbf{x}$ is in $W^\perp$ if and only if $\mathbf{x}$ is orthogonal to every vector in a set that spans W .
2. $W^\perp$ is a subspace of $\mathbb{R}^n$.

Theorem: Let A be an $m \times n$ matrix. The orthogonal complement of the row space of A is the null space of A , and the orthogonal complement of the column space of A is the null space of A^T :

$$(\text{Row } A)^\perp = \text{Nul } A \quad \text{and} \quad (\text{Col } A)^\perp = \text{Nul } A^T$$

Orthogonal Complements

- **Proof:** The row-column rule for computing $A\mathbf{x}$ shows that if $\mathbf{x}$ is in $\text{Nul } A$, then $\mathbf{x}$ is orthogonal to each row of A (with the rows treated as vectors in $\mathbb{R}^n$).
- Since the rows of A span the row space, $\mathbf{x}$ is orthogonal to $\text{Row } A$.
- Conversely, if $\mathbf{x}$ is orthogonal to $\text{Row } A$, then $\mathbf{x}$ is certainly orthogonal to each row of A , and hence $A\mathbf{x} = \mathbf{0}$.
- This proves the first statement of the theorem.

Orthogonal Complements

- Since this statement is true for any matrix, it is true for A^T .
- That is, the orthogonal complement of the row space of A^T is the null space of A^T .
- This proves the second statement, because

$$\text{Row } A^T = \text{Col } A$$

For an $m \times n$ matrix A

Concept	Definition	Dimension	Subset of	Key Properties
Column Space $\text{Col}(A)$	Span of columns of A All linear combinations of columns $\{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^n\}$	$\text{rank}(A)$	$\mathbb{R}^m$	- Image of the transformation - Shows which $\mathbf{b}$ makes $A\mathbf{x} = \mathbf{b}$ solvable - Columns that are pivot columns form a basis
Row Space $\text{Row}(A)$	Span of rows of A Same as $\text{Col}(A^T)$	$\text{rank}(A)$	$\mathbb{R}^n$	- Equal dimension to column space - Non-zero rows of RREF form a basis - Orthogonal to null space
Null Space $\text{Nul}(A)$	Solutions to $A\mathbf{x} = \mathbf{0}$ $\{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$	$\text{nullity}(A)$	$\mathbb{R}^n$	- Kernel of the transformation - Free variables determine dimension - Orthogonal to row space
Left Null Space $\text{Nul}(A^T)$	Solutions to $A^T\mathbf{y} = \mathbf{0}$	$m - \text{rank}(A)$	$\mathbb{R}^m$	- Orthogonal to column space - Related to inconsistent systems

INPUT SPACE ($\mathbb{R}^n$)

OUTPUT SPACE ($\mathbb{R}^m$)



ROW SPACE

"Good Inputs"

$\dim = \text{rank}(A)$

Lives in $\mathbb{R}^n$

$\xrightarrow{A}$



COLUMN SPACE

"Reachable Outputs"

$\dim = \text{rank}(A)$

Lives in $\mathbb{R}^m$



NULL SPACE

"Lost in Translation"

$\dim = \text{nullity}(A)$

Lives in $\mathbb{R}^n$

$\xrightarrow{A} \rightarrow **0** \leftarrow$



LEFT NULL SPACE

"Unreachable Zone"

$\dim = m - \text{rank}(A)$

Lives in $\mathbb{R}^m$

$\perp$

Row Space $\perp$ Null Space

$\perp$

Column Space $\perp$ Left Null Space

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Orthogonal Sets

- A set of vectors $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ in $\mathbb{R}^n$ is said to be an **orthogonal set** if each pair of distinct vectors from the set is orthogonal, that is, if $\mathbf{u}_i \cdot \mathbf{u}_j = 0$ whenever $i \neq j$
- **Theorem:** If $S = \{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ is an orthogonal set of nonzero vectors in $\mathbb{R}^n$, then S is linearly independent and hence is a basis for the subspace spanned by S .

Orthogonal Sets

- **Definition:** An **orthogonal basis** for a subspace W of $\mathbb{R}^n$ is a basis for W that is also an orthogonal set.
- **Theorem:** Let $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ be an orthogonal basis for a subspace W of $\mathbb{R}^n$. For each $\mathbf{y}$ in W , the weights in the linear combination are given by

$$\mathbf{y} = c_1 \mathbf{u}_1 + \cdots + c_p \mathbf{u}_p$$

where

$$c_j = \frac{\mathbf{y} \cdot \mathbf{u}_j}{\mathbf{u}_j \cdot \mathbf{u}_j} \quad (j = 1, \dots, p)$$

An Orthogonal Projection

- Write $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$

where $\hat{\mathbf{y}} = \alpha \mathbf{u}$ for some scalar α and $\mathbf{z}$ is some vector orthogonal to $\mathbf{u}$ (a nonzero vector in $\mathbb{R}^n$).

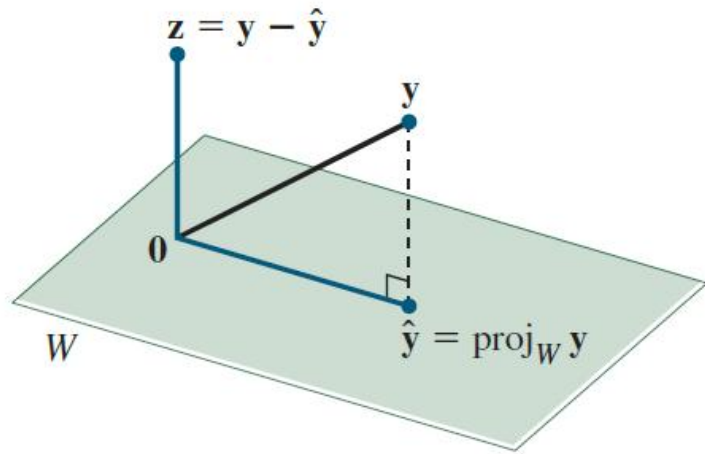


FIGURE 2

Finding α to make $\mathbf{y} - \hat{\mathbf{y}}$ orthogonal to $\mathbf{u}$.

$$\alpha = \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}}$$
$$\hat{\mathbf{y}} = \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u}$$

The vector $\hat{\mathbf{y}}$ is called the **orthogonal projection of $\mathbf{y}$ onto $\mathbf{u}$** , and the vector $\mathbf{z}$ is called the **component of $\mathbf{y}$ orthogonal to $\mathbf{u}$** .

An Orthogonal Projection

- **Example:** Let $\mathbf{y} = \begin{bmatrix} 7 \\ 6 \end{bmatrix}$ and $\mathbf{u} = \begin{bmatrix} 4 \\ 2 \end{bmatrix}$. Find the

orthogonal projection of $\mathbf{y}$ onto $\mathbf{u}$. Then write $\mathbf{y}$ as the sum of two orthogonal vectors, one in $\text{Span}\{\mathbf{u}\}$ and one orthogonal to $\mathbf{u}$.

- **Solution:** Compute

$$\mathbf{y} \cdot \mathbf{u} = 40$$

$$\mathbf{u} \cdot \mathbf{u} = 20$$

An Orthogonal Projection

- The orthogonal projection of $\mathbf{y}$ onto $\mathbf{u}$ is

$$\hat{\mathbf{y}} = \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u} = \begin{bmatrix} 8 \\ 4 \end{bmatrix}$$

and the component of $\mathbf{y}$ orthogonal to $\mathbf{u}$ is

$$\mathbf{y} - \hat{\mathbf{y}} = \begin{bmatrix} 7 \\ 6 \end{bmatrix} - \begin{bmatrix} 8 \\ 4 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}$$

- The sum of these two vectors is $\mathbf{y}$.

An Orthogonal Projection

- That is,

$$\begin{array}{c} \begin{bmatrix} 7 \\ 6 \end{bmatrix} = \begin{bmatrix} 8 \\ 4 \end{bmatrix} + \begin{bmatrix} -1 \\ 2 \end{bmatrix} \\ \uparrow \qquad \qquad \uparrow \qquad \qquad \uparrow \\ y \qquad \qquad \hat{y} \qquad \qquad (y - \hat{y}) \end{array}$$

- The decomposition of $\mathbf{y}$:

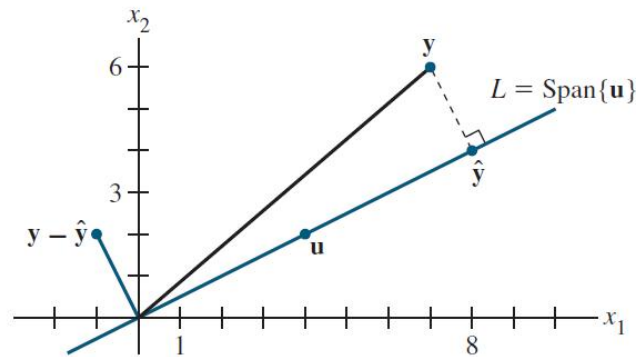


FIGURE 3 The orthogonal projection of $\mathbf{y}$ onto a line L through the origin.

Note: If the calculations above are correct, then $\{\hat{\mathbf{y}}, \mathbf{y} - \hat{\mathbf{y}}\}$ will be an orthogonal set.

Orthogonal Sets

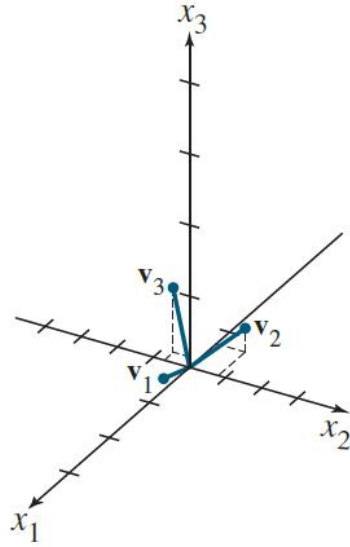
- A set $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ is an **orthonormal set** if it is an orthogonal set of unit vectors.
- If W is the subspace spanned by such a set, then $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ is an **orthonormal basis** for W , since the set is automatically linearly independent.
- The simplest example of an orthonormal set is the standard basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ for $\mathbb{R}^n$. Any nonempty subset of $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is orthonormal.

Orthogonal Sets

- Example: Show that $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is an orthonormal basis of $\mathbb{R}^3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 3/\sqrt{11} \\ 1/\sqrt{11} \\ 1/\sqrt{11} \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} -1/\sqrt{6} \\ 2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} -1/\sqrt{66} \\ -4/\sqrt{66} \\ 7/\sqrt{66} \end{bmatrix}$$

Orthogonal Sets



- When the vectors in an orthogonal set of nonzero vectors are **normalized** to have unit length, the new vectors will still be orthogonal, and hence the new set will be an orthonormal set.

Orthogonal Sets

- **Theorem:** An $m \times n$ matrix U has orthonormal columns if and only if $U^T U = I$.
- **Proof:** To simplify notation, we suppose that U has only three columns, each a vector in $\mathbb{R}^m$.
- Let $U = [\mathbf{u}_1 \quad \mathbf{u}_2 \quad \mathbf{u}_3]$ and compute

$$U^T U = \begin{bmatrix} \mathbf{u}_1^T \\ \mathbf{u}_2^T \\ \mathbf{u}_3^T \end{bmatrix} [\mathbf{u}_1 \quad \mathbf{u}_2 \quad \mathbf{u}_3] = \begin{bmatrix} \mathbf{u}_1^T \mathbf{u}_1 & \mathbf{u}_1^T \mathbf{u}_2 & \mathbf{u}_1^T \mathbf{u}_3 \\ \mathbf{u}_2^T \mathbf{u}_1 & \mathbf{u}_2^T \mathbf{u}_2 & \mathbf{u}_2^T \mathbf{u}_3 \\ \mathbf{u}_3^T \mathbf{u}_1 & \mathbf{u}_3^T \mathbf{u}_2 & \mathbf{u}_3^T \mathbf{u}_3 \end{bmatrix}$$

Orthogonal Sets

- The entries in the matrix at the right are inner products, using transpose notation.
- The columns of U are orthogonal if and only if

$$\mathbf{u}_1^T \mathbf{u}_2 = \mathbf{u}_2^T \mathbf{u}_1 = 0, \quad \mathbf{u}_1^T \mathbf{u}_3 = \mathbf{u}_3^T \mathbf{u}_1 = 0, \quad \mathbf{u}_2^T \mathbf{u}_3 = \mathbf{u}_3^T \mathbf{u}_2 = 0 \quad (4)$$

- The columns of U all have unit length if and only if

$$\mathbf{u}_1^T \mathbf{u}_1 = 1, \quad \mathbf{u}_2^T \mathbf{u}_2 = 1, \quad \mathbf{u}_3^T \mathbf{u}_3 = 1 \quad (5)$$

- The theorem follows immediately from (3)–(5).

Orthogonal Sets

- **Theorem:** Let U be an $m \times n$ matrix with orthonormal columns, and let $\mathbf{x}$ and $\mathbf{y}$ be in $\mathbb{R}^n$.

Then

a. $\|U\mathbf{x}\| = \|\mathbf{x}\|$

b. $U\mathbf{x} \cdot U\mathbf{y} = \mathbf{x} \cdot \mathbf{y}$

c. $U\mathbf{x} \cdot U\mathbf{y} = 0$ if and only if $\mathbf{x} \cdot \mathbf{y} = 0$

- Properties (a) and (c) say that the linear mapping preserves lengths and orthogonality.

$$\mathbf{x} \mapsto U\mathbf{x}$$

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Section 6.2: Orthogonal Sets

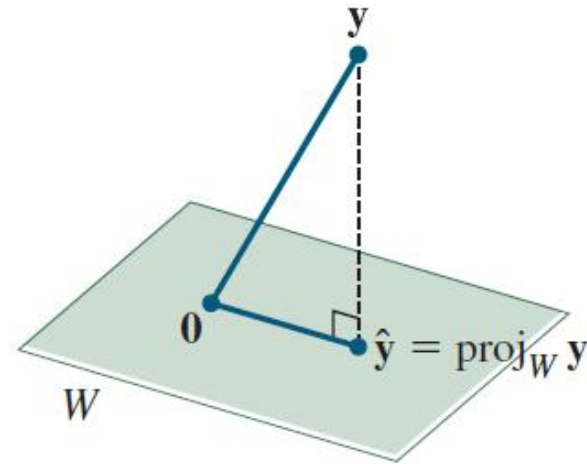
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Orthogonal Projections

- Given a vector $\mathbf{y}$ and a subspace W in $\mathbb{R}^2$, there is a vector $\hat{\mathbf{y}}$ in W such that
 - a. $\hat{\mathbf{y}}$ is the unique vector in W for which $\mathbf{y} - \hat{\mathbf{y}}$ is orthogonal to W , and
 - b. $\hat{\mathbf{y}}$ is the unique vector in W closest to $\mathbf{y}$.



These two properties of $\hat{\mathbf{y}}$ provide the key to finding the least-squares solutions of linear systems.

The Orthogonal Decomposition Theorem

- **Theorem:** Let W be a subspace of $\mathbb{R}^n$. Then each $\mathbf{y}$ in $\mathbb{R}^n$ can be written uniquely in the form

$$\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z} \quad (1)$$

where $\hat{\mathbf{y}}$ is in W and $\mathbf{z}$ is in $W^\perp$.

- In fact, if $\{\mathbf{u}_1, \dots, \mathbf{u}_p\}$ is any orthogonal basis of W , then

$$\hat{\mathbf{y}} = \frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 + \dots + \frac{\mathbf{y} \cdot \mathbf{u}_p}{\mathbf{u}_p \cdot \mathbf{u}_p} \mathbf{u}_p \quad (2)$$

and $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}}$.

The Orthogonal Decomposition Theorem

• **Example:** Let $\mathbf{u}_1 = \begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix}$, $\mathbf{u}_2 = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$, and $\mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$.

Observe that $\{\mathbf{u}_1, \mathbf{u}_2\}$ is an orthogonal basis for $W = \text{Span}\{\mathbf{u}_1, \mathbf{u}_2\}$.
Write $\mathbf{y}$ as the sum of a vector in W and a vector orthogonal to W .

The Orthogonal Decomposition Theorem

- **Solution:** The orthogonal projection of $\mathbf{y}$ onto W is

$$\hat{\mathbf{y}} = \frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \mathbf{u}_1 + \frac{\mathbf{y} \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2} \mathbf{u}_2 = \begin{bmatrix} -2/5 \\ 2 \\ 1/5 \end{bmatrix}$$

Also

$$\mathbf{y} - \hat{\mathbf{y}} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} - \begin{bmatrix} -2/5 \\ 2 \\ 1/5 \end{bmatrix} = \begin{bmatrix} 7/5 \\ 0 \\ 14/5 \end{bmatrix}$$

The Orthogonal Decomposition Theorem

- The desired decomposition of $\mathbf{y}$ is

$$\mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -2/5 \\ 2 \\ 1/5 \end{bmatrix} + \begin{bmatrix} 7/5 \\ 0 \\ 14/5 \end{bmatrix}$$

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The Gram-Schmidt Process

- **Theorem: The Gram-Schmidt Process**

- Given a basis $\{\mathbf{x}_1, \dots, \mathbf{x}_p\}$ for a nonzero subspace W of $\mathbb{R}^n$, define

$$\mathbf{v}_1 = \mathbf{x}_1$$

$$\mathbf{v}_2 = \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1$$

$$\mathbf{v}_3 = \mathbf{x}_3 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2} \mathbf{v}_2$$
$$\vdots$$

$$\mathbf{v}_p = \mathbf{x}_p - \frac{\mathbf{x}_p \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1 - \frac{\mathbf{x}_p \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_2} \mathbf{v}_2 - \dots - \frac{\mathbf{x}_p \cdot \mathbf{v}_{p-1}}{\mathbf{v}_{p-1} \cdot \mathbf{v}_{p-1}} \mathbf{v}_{p-1}$$

- Then $\{\mathbf{v}_1, \dots, \mathbf{v}_p\}$ is an orthogonal basis for W . In addition
 $\text{Span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = \text{Span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$ for $1 \leq k \leq p$. (1)

Orthonormal Bases

- **Example** For the orthogonal basis

$$\mathbf{v}_1 = \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

- An orthonormal basis is

$$\mathbf{u}_1 = \frac{1}{\|\mathbf{v}_1\|} \mathbf{v}_1 = \frac{1}{\sqrt{45}} \begin{bmatrix} 3 \\ 6 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \\ 0 \end{bmatrix}$$
$$\mathbf{u}_2 = \frac{1}{\|\mathbf{v}_2\|} \mathbf{v}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

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Least-Squares Problems

- **Definition:** If A is $m \times n$ and $\mathbf{b}$ is in $\mathbb{R}^m$, a **least-squares solution** of $A\mathbf{x} = \mathbf{b}$ is an $\hat{\mathbf{x}}$ in $\mathbb{R}^n$ such that

$$\|\mathbf{b} - A\hat{\mathbf{x}}\| \leq \|\mathbf{b} - A\mathbf{x}\|$$

for all $\mathbf{x}$ in $\mathbb{R}^n$.

- So we seek an $\mathbf{x}$ that makes $A\mathbf{x}$ the closest point in $\text{Col } A$ to $\mathbf{b}$.

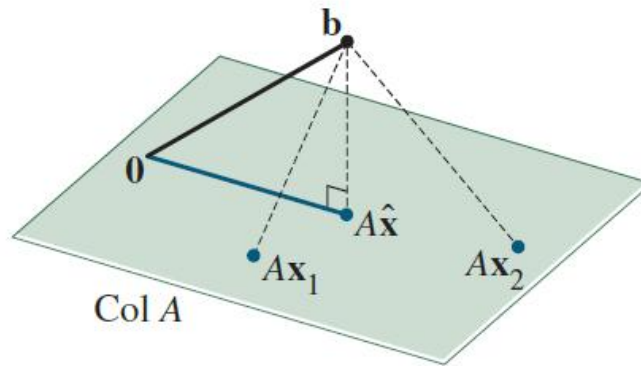


FIGURE 1 The vector $\mathbf{b}$ is closer to $A\hat{\mathbf{x}}$ than to $A\mathbf{x}$ for other $\mathbf{x}$.

Solution of the General Least-Squares Problems

- **Solution of the General Least-Squares Problem**

Let $\hat{\mathbf{b}} = \text{proj}_{\text{Col } A} \mathbf{b}$, there is an $\hat{\mathbf{x}}$ in $\mathbb{R}^n$ such that $A\hat{\mathbf{x}} = \hat{\mathbf{b}}$

- Since $\hat{\mathbf{b}}$ is the closest point in $\text{Col } A$ to $\mathbf{b}$, a vector $\hat{\mathbf{x}}$ is a least-squares solution of $A\mathbf{x} = \mathbf{b}$ if and only if $\hat{\mathbf{b}}$ satisfies $A\hat{\mathbf{x}} = \hat{\mathbf{b}}$.

- Such an $\hat{\mathbf{x}}$ in $\mathbb{R}^n$ is a list of weights that will build $\hat{\mathbf{b}}$ out of the columns of A

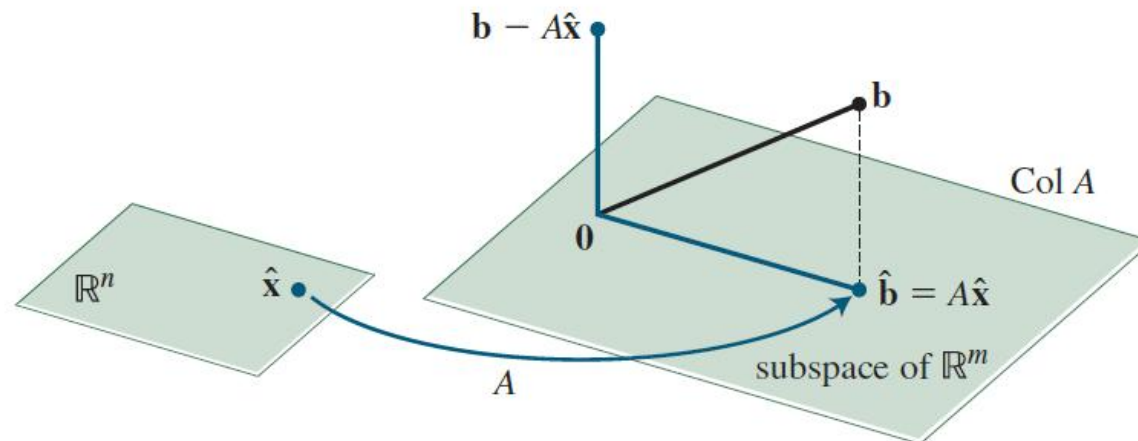


FIGURE 2 The least-squares solution $\hat{\mathbf{x}}$ is in $\mathbb{R}^n$.

Solution of the General Least-Squares Problems

- Suppose $\hat{\mathbf{x}}$ satisfies $A\hat{\mathbf{x}} = \hat{\mathbf{b}}$.
- By the Orthogonal Decomposition Theorem, the projection $\hat{\mathbf{b}}$ has the property that $\mathbf{b} - \hat{\mathbf{b}}$ is orthogonal to $\text{Col } A$, so $\mathbf{b} - A\hat{\mathbf{x}}$ is orthogonal to each column of A .
- If $\mathbf{a}_j$ is any column of A , then $\mathbf{a}_j \cdot (\mathbf{b} - A\hat{\mathbf{x}}) = 0$, so $\mathbf{a}_j^T (\mathbf{b} - A\hat{\mathbf{x}}) = 0$.

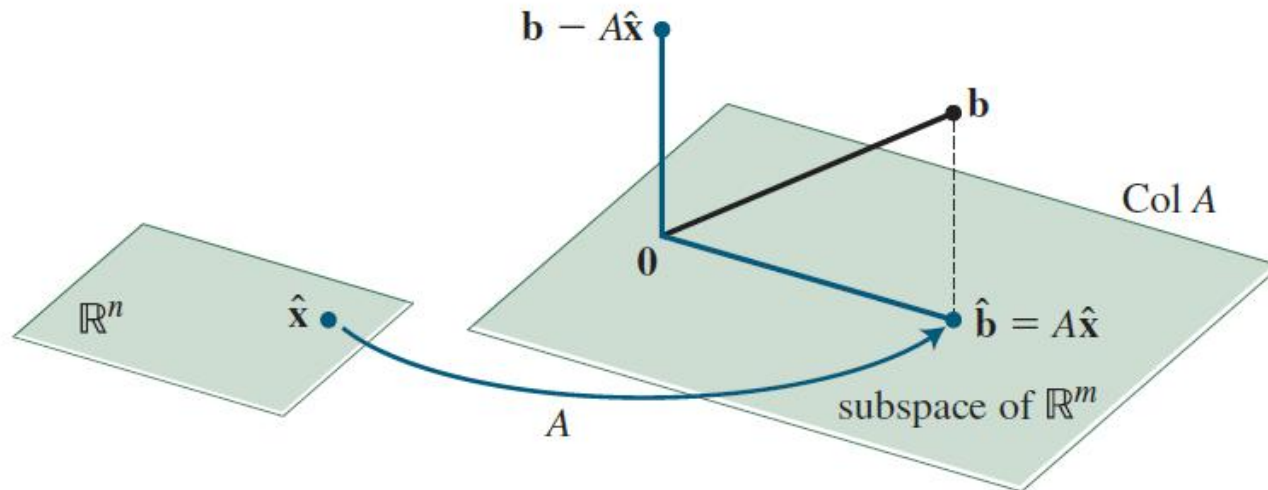


FIGURE 2 The least-squares solution $\hat{\mathbf{x}}$ is in $\mathbb{R}^n$.

Solution of the General Least-Squares Problems

- Since each $\mathbf{a}_j^T$ is a row of A^T ,

$$A^T (\mathbf{b} - A\hat{\mathbf{x}}) = 0 \quad (2)$$

Thus

$$A^T \mathbf{b} - A^T A\hat{\mathbf{x}} = 0$$

$$A^T A\hat{\mathbf{x}} = A^T \mathbf{b}$$

These calculations show that each least-squares solution of $A\mathbf{x} = \mathbf{b}$ satisfies the equation $A^T A\mathbf{x} = A^T \mathbf{b}$. (3)

The matrix equation (3) represents a system of equations called the **normal equations** for $A\mathbf{x} = \mathbf{b}$.

- A solution of (3) is often denoted by $\hat{\mathbf{x}}$.

Solution of the General Least-Squares Problems

- **Theorem:** The set of least-squares solutions of $A\mathbf{x} = \mathbf{b}$ coincides with the nonempty set of solutions of the normal equation $A^T A\mathbf{x} = A^T \mathbf{b}$.

Solution of the General Least-Squares Problems

- **Example:** Find a least-squares solution of the inconsistent system $A\mathbf{x} = \mathbf{b}$ for

$$A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix}$$

- **Solution:** To use normal equations (3), compute:

$$A^T A = \begin{bmatrix} 4 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 17 & 1 \\ 1 & 5 \end{bmatrix}$$

Solution of the General Least-Squares Problems

$$A^T \mathbf{b} = \begin{bmatrix} 4 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix} = \begin{bmatrix} 19 \\ 11 \end{bmatrix}$$

- Then the equation $A^T A \mathbf{x} = A^T \mathbf{b}$ becomes

$$\begin{bmatrix} 17 & 1 \\ 1 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 19 \\ 11 \end{bmatrix}$$

Solution of the General Least-Squares Problems

- Row operations can be used to solve the system on the previous slide, but since $A^T A$ is invertible and 2×2 , it is probably faster to compute

$$(A^T A)^{-1} = \frac{1}{84} \begin{bmatrix} 5 & -1 \\ -1 & 17 \end{bmatrix}$$

and then solve $A^T A \mathbf{x} = A^T \mathbf{b}$ as

$$\begin{aligned} \hat{\mathbf{x}} &= (A^T A)^{-1} A^T \mathbf{b} \\ &= \frac{1}{84} \begin{bmatrix} 5 & -1 \\ -1 & 17 \end{bmatrix} \begin{bmatrix} 19 \\ 11 \end{bmatrix} = \frac{1}{84} \begin{bmatrix} 84 \\ 168 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \end{aligned}$$